



[The exam text is in English. You should answer the questions in English.]

The exam consists of 13 questions (6 pages). Please first check if any page is missing in the printout you. For the open questions: please keep the handwriting clear and readable.

The exam starts at 9.00 a.m. and ends at 12.00 p.m. Participation in the exam requires being present for **at least 1 hour**.

Electronic devices are not allowed and must be stowed.
Points for each open question are indicated.

- **Scrap paper will NOT be evaluated. Please write all your answers on these sheets!**
- *Please write readable and write your name and student identification number on every single sheet of paper for unique identification!*
- **Don't spend much time writing long stories – that's NOT necessary. Don't panic! You CAN make it!**
- *Look at the whole exam first to get an overview, pay attention to the grading scheme of multiple-choice questions, and do those assignments first which you feel are doable for you.*
- *Good success and all the best for your studies!*

Part I - Multiple-choice [50 points].

- Each problem has a variable number of correct statements. Please write your answers in the table given at the end of the question sheet.
- In NO problem, all statements are correct. If you choose all the options, i.e., "ABCD", the score for that question will be ZERO!
- The score for each question will be given according to Morgan method:

$$5 \times \max \{ r / \#correct\ option(s) - w / \#distractors(s), 0 \},$$

where:

- 5 is the total score for each question;
- r is the number of option(s) you select correctly;
- w is the number of option(s) you select incorrectly;
- $\#correct\ option(s)$ is the number of correct options of this question;
- $\#distractors(s)$ is the number of incorrect options/distractors of this question;
- Please DO NOT mark an option if you are not certain with it as a penalty will incur.

Please write the alphabetic characters (e.g., CD) representing your answers of the multiple-choice questions into the table below.

No.	1	2	3	4	5	6	7	8	9	10
Answers										

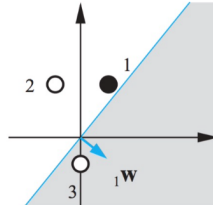
1. Regarding the biological neuron and the McCulloch-Pitts model, which of the following descriptions are correct? [more than one answer is correct]
 - a. Dendrites of a biological neuron receive incoming signals, similar to the input nodes in the McCulloch-Pitts model.
 - b. The axon of a biological neuron transmits signals to other neurons, which is analogous to the bias term in the McCulloch-Pitts model.
 - c. Synaptic strength of the biological neuron is represented by the weights in the McCulloch-Pitts model.
 - d. Spikes, or action potentials, in a biological neuron are analogous to the output of the step function in the McCulloch-Pitts model.
2. What statements are true about the stability condition for the Hopfield Network? [more than one answer is correct]
 - a. A pattern is an attractor of the network if the weights on the edges are inversely proportional to the product of the node states.
 - b. The condition for a stored pattern to be stable is that the node state of the evolving pattern is equal the node state of the stored pattern, for each node.
 - c. Stability is obtained if $w_{ij} = \frac{1}{N} \xi_i \xi_j$.
 - d. Being a stable pattern and being an attractor for a network cannot be coexisting conditions.
3. Which of the following statements is true about Hebbian learning? [one answer is correct]
 - a. Hebb's rule says that the weight between two neurons should be decreased if they are activated simultaneously.
 - b. The learning rate in Hebbian learning is usually larger than one.
 - c. The Hebbian learning process will always stabilize after passing the training set several times.
 - d. None of the above.
4. Which of the following statements are true about multi-layer perceptron (MLP)? [more than one answer is correct]
 - a. MLPs consist of an input layer, one or more hidden layers, and an output layer.
 - b. Each neuron in an MLP applies a linear activation function to its inputs.
 - c. MLPs are capable of approximating any continuous function given sufficient neurons and layers.
 - d. MLPs are inherently able to handle sequential data without modification.
5. Which of the following statements are true regarding the backpropagation algorithm? [one answer is correct]
 - a. Back-propagation updates weights by minimizing the output values directly.
 - b. Back-propagation requires labeled data to compute the loss function.
 - c. The back-propagation algorithm operates only on feedforward neural networks.
 - d. Back-propagation can be used without computing gradients.
6. Which of the following statements are true regarding the long short-term memory networks? [more than one answer is correct]
 - a. Use an extended version of the back-propagation algorithm for training.
 - b. Use the ReLU activation function.
 - c. The functionality of the cell state is to allow vanishing gradients.
 - d. LSTMs leverage gating mechanisms (input gate, forget gate, output gate) to control the flow of information and gradients.



7. Which of the following statements are true regarding RNNs? [more than one answer is correct]
 - a. RNNs are capable of memorizing data.
 - b. RNNs do not contain cycles in the network.
 - c. The output of a RNN is a constant given a constant input.
 - d. Hidden states summarize all inputs a RNN receives in the history.
8. Which of the following tasks would be more suitable to use convolution operation? [one answer is correct]
 - a. Forecast the probability of raining, provide a weather history.
 - b. Classify images containing dogs from cats.
 - c. Solving the XOR problem.
 - d. Generate text from images (figure captioning)
9. Which of the following statements are true about the image data? [more than one answer is correct]
 - a. Image data are usually represented by tensors/matrix.
 - b. Colored image input also contains three (RGB) channels.
 - c. Flattening the image incurs no information loss.
 - d. Performing pooling operation on images incurs information loss.
10. Please calculate the size of the output image (assuming the stride is two) for a given input image and kernel size. Case (1): input size 8x8, kernel size 4x4; Case (2): input size 15x15, kernel size 5x5. [more than one answer is correct]
 - a. For case 1, if no padding, the size of the output matrix is 3x3.
 - b. For case 2, if no padding, the size of the output volume is 6x6.
 - c. For case 1, if the padding size is (1, 1), the size of the output volume is 4x4.
 - d. For case 2, if the padding size is (2, 2), the size of the output volume is 7x7.

Part II - Open Questions [50 points]

1. (15 points) Consider a perceptron $O = \text{sgn}(\vec{w}^T \vec{x})$ (without the bias term) and binary classification task depicted below [target label: black = 1, white = -1].



- a. (5) Now we wish to train this perceptron with Hebb's rule. Please write down Hebb's rule mathematically [hint: don't forget the learning rate]
- b. (5) Imagine the initial weight and the decision boundary are set according to the above diagram. Note that the shaded half space is classified to the **black** label. Please demonstrate the Hebb's rule visually in this diagram by considering **point #3** [please draw a new figure below].
- c. (5) Now consider the famous eXclusive OR (XOR) problem. Can this problem be solved by a perceptron theoretically? Please picture this XOR problem and argue in this picture why this problem can/cannot be solved by a simple perceptron.



- b. **(10)** Prove the stability condition for a pattern ξ^v .

- c. **(5)** Please define the crosstalk term.

3. **(15 points)** Consider the convolution and pooling operations and answer the following questions:
- a. **(5)** Compute the resulting feature map using a stride of 2 and no padding.

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 1 & 2 \\ 4 & 1 & 0 & 1 & 7 & 3 \\ 1 & 3 & 2 & 2 & 3 & 0 \\ 0 & 1 & 3 & 1 & 1 & 5 \\ 2 & 4 & 1 & 2 & 0 & 1 \\ 1 & 2 & 1 & 3 & 1 & 4 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} =$$

- b. **(5)** Given the same input matrix, compute the resulting feature map using a min pooling operation with a pool size 2x2.
- c. **(5)** Please draw the CNN feature learning + classification pipeline and give a brief description for the two phases.